

1 Chapter 1

Nick's Question 1.1 Proof of lemma 1.8.4

The proof is very nice.

2 Chapter 2

Nick's Question 2.1 In the proof of Theorem 2.4.3 on the bottom of page 34 there are some claims made that I am unable to verify why they are true. First, we are considering

$$X_t = X_0 + \int_0^t u(s)dB_s$$

and here $u(s)$ only satisfies

$$\mathbb{P}\left(\int_0^t u(s)^2 ds < \infty\right) = 1$$

so, as in the Kuo text, this stochastic integral exists and a limit in probability of a sequence of stochastic integrals $\int_0^t u_n(s)dB_s$ where the u_n are simple and are in L^2_{ad} and hence $\int_0^t u_n(s)dB_s$ is a martingale and we know the first and second moment. But for the limit $\int_0^t u(s)dB_s$ we know nothing about the moments. But in this text, it is claimed that

$$\mathbb{E}\left[\int_{t_i}^{t_{i+1}} u^2(s)ds - \left(\int_{t_i}^{t_{i+1}} u(s)dB_s\right)^2 \middle| \mathcal{F}_i\right] = 0$$

and I have never seen this claimed in any other text.

Note that u is bounded (we are assuming localization has been done). Hence, all moments exists.

Nick's Question 2.2 In the proof of theorem 2.5.1, why is it true that

$$E\left(\int_0^t |u'_n(B_s) - 1_{(x,\infty)}(B_s)|^2 ds\right) \leq \int_0^t \mathbb{P}\left(|B_s - x| \leq \frac{2}{n}\right) ds$$

Please explore the mollifier f and f_n . Plot these u'_n and see how could it differ from $1_{(x,\infty)}$.

Nick's Question 2.3 What is the motivation on page 41 behind the following definition of the backward ito integral

$$\int_0^T u(s)\hat{d}B_s = \lim_{n \rightarrow \infty} \sum_{i=0}^{n-2} \left(\frac{n}{T} \int_{(j+1)T/n}^{(j+2)T/n} u(s)ds\right) (B_{(j+1)T/n} - B_{jT/n})$$

The integral in the parenthesis is the average of u_s on $((j+1)T/n, (j+2)T/n)$.

Let's skip this part. I know nothing on the backward SDE and not particular interested either. It has applications in finance. If you don't do finance, I don't see the point here.

Nick's Question 2.4 (Problem 2.7)

This problem follows easily from Levy's Characterization of Brownian Motion. But I can't figure out how to do this problem with the hint given.

This is a nice problem. The hints is to ask you to find the distribution of $X_t - X_s$ through the characteristic function. Once apply Ito's formula, take expectation and remove the martingale part, you should arrive at an integral equation. Take derivatives, you arrive at a differential equation. The solution for $a_s(t) := \mathbb{E}(e^{i\lambda(X_t - X_s)})$ is

$$a_s(t) = e^{-\frac{\lambda^2}{2}(t-s)}.$$

Nick's Question 2.5 (Problem 2.15)

First we consider the 2 - d Ito process

$$\begin{pmatrix} M_s \\ N_s \end{pmatrix} = \begin{pmatrix} E(X^p) \\ E(Y^{q'}) \end{pmatrix} + \int_0^t \begin{pmatrix} \varphi_s & 0 \\ 0 & \psi_s \end{pmatrix} \begin{pmatrix} d\hat{B}_s \\ dB_s \end{pmatrix}$$

and then following the hint we apply Ito's formula to $f(x, y) = x^{1/p}y^{1/q'}$. We get the following derivatives:

- $f_x = \frac{1}{p}x^{\frac{1}{p}-1}y^{\frac{1}{q'}}$
- $f_{xx} = \frac{1}{p}\left(\frac{1}{p}-1\right)x^{\frac{1}{p}-2}y^{\frac{1}{q'}}$
- $f_y = \frac{1}{q'}x^{\frac{1}{p}}y^{\frac{1}{q'}-1}$
- $f_{yy} = \frac{1}{q'}\left(\frac{1}{q'}-1\right)x^{\frac{1}{p}}y^{\frac{1}{q'}-2}$

Now putting this into Ito's Formula gives us

$$\begin{aligned} M_t^{1/p} N_t^{1/q'} &= \|X\|_p \|Y\|_{q'} + \frac{1}{p} \int_0^t M_s^{\frac{1}{p}-1} N_s^{\frac{1}{q'}} \varphi_s d\hat{B}_s + \frac{1}{q'} \int_0^t M_s^{\frac{1}{p}} N_s^{\frac{1}{q'}-1} \psi_s dB_s \\ &\quad + \frac{1}{2} \frac{1}{p} \left(\frac{1}{p}-1\right) \int_0^t M_s^{\frac{1}{p}-2} N_s^{\frac{1}{q'}} \varphi_s^2 ds + \frac{1}{2} \frac{1}{q'} \left(\frac{1}{q'}-1\right) \int_0^t M_s^{\frac{1}{p}} N_s^{\frac{1}{q'}-2} \psi_s^2 ds \end{aligned}$$

I am not really sure where to continue from here. I see that if we let $t \rightarrow \infty$ then $M_t^{1/p} N_t^{1/q'} \rightarrow XY$ so if we took the expected value of both sides then we would partially have what we are looking for since $E(XY)$ is on the left and $\|X\|_p \|Y\|_{q'}$ is on the right. But I am not sure about the other terms. Im guessing the Ito integrals in the right have 0 expectation but I am not sure if that is in fact true since I dont see how it is easily seen that the integrands are in $L^2(\mathcal{P})$.

What I did next was see what $\hat{B}_s$ was equal too and for this we define another Ito Process

$$\begin{pmatrix} B_s \\ \tilde{B}_s \end{pmatrix} = \int_0^t \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} dB_s \\ d\tilde{B}_s \end{pmatrix}$$

and applied Itos formula to $f(t, x, y) = e^{-t}x + \sqrt{1 - e^{-2t}}y$ which gives us

$$\hat{B}_s = e^{-t}B_s + \sqrt{1 - e^{-2t}}\tilde{B}_s = \int_0^t \frac{e^{-2s}}{\sqrt{1 - e^{-2s}}}\tilde{B}_s - e^{-s}B_s ds + \int_0^t e^{-s}dB_s + \int_0^t \sqrt{1 - e^{-2s}}d\tilde{B}_s$$

However I am not sure if this step is necessary.

The followings are some of my thoughts: First of all, we need the absolute values:

$$\begin{pmatrix} M_s \\ N_s \end{pmatrix} = \begin{pmatrix} \mathbb{E}(|X|^p) \\ \mathbb{E}(|Y|^{q'}) \end{pmatrix} + \int_0^t \begin{pmatrix} \varphi_s & 0 \\ 0 & \psi_s \end{pmatrix} \begin{pmatrix} d\hat{B}_s \\ dB_s \end{pmatrix}.$$

Second, you forget

$$f_{xy} = \frac{1}{p} \frac{1}{q'} x^{-1/(p')} y^{-1/q'}.$$

Hence, by Ito's formula, we will have

$$\begin{aligned} M_t^{1/p} N_t^{1/q'} &= \|X\|_p \|Y\|_{q'} + \frac{1}{p} \int_0^t M_s^{\frac{1}{p}-1} N_s^{\frac{1}{q'}} \varphi_s d\hat{B}_s + \frac{1}{q'} \int_0^t M_s^{\frac{1}{p}} N_s^{\frac{1}{q'}-1} \psi_s dB_s \\ &\quad + \frac{1}{2} \frac{1}{p} \left(\frac{1}{p} - 1 \right) \int_0^t M_s^{\frac{1}{p}-2} N_s^{\frac{1}{q'}} \varphi_s^2 ds + \frac{1}{2} \frac{1}{q'} \left(\frac{1}{q'} - 1 \right) \int_0^t M_s^{\frac{1}{p}} N_s^{\frac{1}{q'}-2} \psi_s^2 ds \\ &\quad + \frac{1}{pq'} \int_0^t M_s^{-\frac{1}{p'}} N_s^{-\frac{1}{q}} \psi_s \phi_s d\langle B, \hat{B} \rangle_s \\ &=: \|X\|_p \|Y\|_{q'} + \sum_{i=1}^5 I_i. \end{aligned}$$

Upon taking expectation on both sides, we see that

$$LHS = \mathbb{E}(|X_t Y_t|) \geq \mathbb{E}(X_t Y_t).$$

We only need to prove that

$$\mathbb{E} \left(\sum_{i=1}^5 I_i \right) \leq 0,$$

which is clear for $i = 1, 2$, in which case we have the expectation equal to zero. Now for the rest three parts, we see that

$$I_3 + I_4 + I_5 = -\frac{1}{2pq'} M_s^{\frac{1}{p}-2} N_s^{\frac{1}{q'}-2} \left(N_s^2 \varphi_s^2 \frac{q'}{p'} - 2M_s N_s \phi_s \psi_s e^{-s} + M_s \psi_s^2 \frac{p}{q} \right),$$

where we have used the fact that $d\langle B, \hat{B} \rangle_s = e^{-s} ds$. Hence, the condition become the condition that the quadratic form has no solution:

$$\Delta = 4e^{-2s} - 4\frac{q'p}{p'q} = 4 \left(e^{-2s} - \frac{p-1}{q-1} \right) \leq 0$$

which is equivalent to

$$q \leq e^{2s}(p-1) + 1.$$

Hence, the largest q that makes the inequality work is $q = e^{2s}(p-1) + 1$.

3 Chapter 3

3.1 Reading Material

Nick's Question 3.1 (Smooth Cylindrical Random Variable) Following chapter 3 of the Kuo text, denote C the Wiener space of continuous functions from $[0,1]$ to $\mathbb{R}$, $\mathcal{R}$ the collection cylindrical sets

$$A = \{\omega \in C : (\omega(t_1), \dots, \omega(t_n)) \in U \mid t_i \in [0, 1], n \in \mathbb{N}, U \in \mathcal{B}(\mathbb{R}^n)\}$$

and μ the associated Wiener measure. Consider the coordinate processes $B(t, \omega) = \omega(t)$ which are Brownian motions under $(C, \mathcal{B}(\mathcal{R}), \mu)$. There is a theorem that can be found [here](#).

Theorem : The random variables

$$\{f(B(t_1, \cdot), \dots, B(t_n, \cdot)) \mid n \in \mathbb{N}, f \in \mathcal{S}(\mathbb{R}^n)\}$$

are dense in $L^2(C, \mu)$ where $\mathcal{S}(\mathbb{R}^n)$ are the Schwartz functions.

How does this relate to $\mathcal{S}$ defined on page 51 of the intro to Malliavin textbook? How are the cylindrical random variables defined in this case? Why is $\mathcal{S}$ dense in $L^2(\Omega)$? Even in Diffusions Markov Processes and Martingales - Roger and Williams, they define the cylindrical sigma field in the same way Kuo does- through the use of coordinate processes.

Nick Solution.

density

Theorem 3.1. Consider the probability space $(\Omega, \mathcal{F}, P)$ where $\mathcal{F}$ is the standard Brownian filtration. Then $\{f(B(h_1), \dots, f(B(h_n)), n \in \mathbb{N}, f \in C_p^\infty(\mathbb{R}^n), h_i \in H := L^2(\mathbb{R}^+)\}$ is dense in $L^2(\Omega)$.

Proof: This proof follows the same scheme as the solution found [here](#) which utilizes Dynkin's multiplicative system theorem. First, we can easily show that the sigma field generated by $\{B(h)\}_{h \in H}$ generates $\mathcal{F}$. This is true since $1_{[0,t]} \in H$ and $B(1_{[0,t]}) = B_t$. Second, since $\mathcal{S}(\mathbb{R}) \subset C_p^\infty(\mathbb{R})$ and since for every $h \in H$ there exists functions $f_n \in \mathcal{S}(\mathbb{R})$ such that $f_n \rightarrow h$ in $L^2(\mathbb{R})$ then by the Wiener Isometry we have $E|\int_0^\infty h(s) - f_n(s)dB_s|^2 = \int_0^\infty |h(s) - f_n(s)|^2 ds \rightarrow 0$. This implies that the sigma field generated by $\{f(B(h))\}$ where $h \in H$ and $f \in C_p^\infty(\mathbb{R})$ is the same as $\mathcal{F}$.

Now we let $M = \{f(B(h_1), \dots, B(h_n))\}$. We have already shown that $\sigma(M)$ is the same as $\mathcal{F}$. Also, M is closed under pointwise multiplication. Let $E := \bar{M}$ where closure is with respect to the $L^2(\Omega)$ norm. Let $H := E \cap L^\infty(\Omega)$. We have $M \subset H$. It can be shown that H is closed with respect to bounded pointwise convergence and also that H contains all constant functions. Thus, the hypothesis is Dynkin's multiplicative theorem has been satisfied and this implies that H contains all $\sigma(M) = \mathcal{F}$ measurable functions. But this means that it contains all bounded Borel measurable functions. But by the definition of H , this implies that E contains all the bounded Borel measurable functions. We know that all $X \in L^2(\Omega)$ can be approximated in the $L^2(\Omega)$ norm by bounded measurable functions. Also since E is closed and contains the bounded measurable functions, then E itself must be equal to $L^2(\Omega)$. Therefore we conclude that $\bar{M} = E = L^2(\Omega)$. □

Nick's Question 3.2 Why is $(D_t F)_{t \geq 0}$ a stochastic processes in $L^2(\Omega; H)$ where $H = L^2(\mathbb{R}^+)$? $D_t F$ is defined as

$$D_t F = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(B(h_1), \dots, B(h_n)) h_i(t)$$

where $h_i \in H$ and $B(h_i) = \int_0^\infty h_i dB_s$ is the standard Wiener Integral. So isn't $D_t F$ real

valued since we are evaluating the $h_i(\cdot)$ at t ? Then shouldn't the $(D_t F)_{t \geq 0}$ be a $L^2(\Omega; \mathbb{R})$ valued stochastic process?

I believe that this is just a typo in the text

Nick's Question 3.3 (Proposition 3.2.4)(All questions have been solved with work below)

1. Why does $D_h(\delta(u)) = \langle u, h \rangle + \delta(D_h u)$? (I have answered this below)
2. Also, at the bottom of page 52, why is it that (I have answered this below)

$$E \left(\sum_{i,j=1}^{\infty} D_{e_i} \langle u, e_j \rangle D_{e_j} \langle v, e_i \rangle \right) = E \left(\int_0^{\infty} \int_0^{\infty} D_s u_t D_t v_s ds dt \right)$$

3. In the proof of property 3.3 why do we have (Answered below)

$$E \left(\sum_{i=1}^{\infty} \langle v, e_i \rangle \left(\langle u, e_i \rangle + \delta(D_{e_i}(u)) \right) \right) = E(\langle u, v \rangle) + E \left(\sum_{i,j=1}^{\infty} D_{e_i} \langle e, e_j \rangle D_{e_j} \langle v, e_i \rangle \right)$$

Proof of $D_h(\delta(u)) = \langle u, h \rangle + \delta(D_h u)$.

Since $u = \sum F_j h_j$ we just apply the definition of δ to get

$$\begin{aligned} D_h(\delta(u)) &= D_h \left(\sum_{j=1}^n F_j B(h_j) - \sum_{j=1}^n \langle DF_j, h_j \rangle \right) \\ &= \sum_{j=1}^n D_h(F_j B(h_j)) - \sum_{j=1}^n D_h \langle DF_j, h_j \rangle \\ &= \sum_{j=1}^n F_j \langle h, h_j \rangle + \sum_{j=1}^n B(h_j) D_h(F_j) - \sum_{j=1}^n D_h \langle DF_j, h_j \rangle \end{aligned} \quad (3.1) \quad \boxed{\text{P:3.2.4-1}}$$

where the third inequality follows from the product rule and $D_h(B(h_j)) = \langle h, h_j \rangle$. Now we focus on the last term from the last line above (3.1).

$$\sum_{j=1}^n D_h \langle DF_j, h_j \rangle = \sum_{j=1}^n \langle D_t \langle DF_j, h_j \rangle, h(t) \rangle \quad (3.2) \quad \boxed{\text{P:3.2.4-6}}$$

Before we continue, we simplify $D_t \langle DF_j, h_j \rangle$. We may assume that $F_j = f^j(B(h_1^j(t)), \dots, B(h_n^j(t)))$ where $f^j \in C_p^\infty(\mathbb{R}^n)$ and $h_i^j \in H$.

$$\begin{aligned} D_t \langle DF_j, h_j \rangle &= D_t \left\langle \sum_{i=1}^n \frac{\partial f^j}{\partial x_i} h_i^j(s), h_j(s) \right\rangle \\ &= \sum_{i=1}^n \langle h_i^j(s), h_j(s) \rangle D_t \left(\frac{\partial f^j}{\partial x_i} \right) \end{aligned} \quad (3.3) \quad \boxed{\text{P:3.2.4-2}}$$

We now plug (3.3) into (3.2) above to get

$$\sum_{j=1}^n \langle D_t \langle DF_j, h_j \rangle, h(t) \rangle = \sum_{j=1}^n \sum_{i=1}^n \langle h_i^j(s), h_j(s) \rangle \langle D_t \left(\frac{\partial f^j}{\partial x_i} \right), h(t) \rangle$$

$$\begin{aligned}
&= \sum_{j=1}^n \sum_{i=1}^n \left\langle h_i^j(s) \left\langle D_t \left(\frac{\partial f^j}{\partial x_i} \right), h(t) \right\rangle, h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle \left\langle D_t \left(\sum_{i=1}^n h_i^j(s) \frac{\partial f^j}{\partial x_i} \right), h(t) \right\rangle, h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle \left\langle D_t (DF_j), h(t) \right\rangle, h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle D_h (DF_j), h_j(s) \right\rangle
\end{aligned}$$

and if I haven't made a mistake, then what we have shown so far is that

$$\begin{aligned}
D_h(\delta(u)) &= \sum_{j=1}^n F_j \langle h, h_j \rangle + \sum_{j=1}^n B(h_j) D_h(F_j) - \sum_{j=1}^n D_h \langle DF_j, h_j \rangle \\
&= \sum_{j=1}^n F_j \langle h, h_j \rangle + \sum_{j=1}^n B(h_j) D_h(F_j) - \sum_{j=1}^n \left\langle D_h (DF_j), h_j \right\rangle
\end{aligned} \tag{3.4} \quad \boxed{\text{P:3.2.4-3}}$$

Also, the above is exactly line two of the first centered equations on page 53.

Now we need to show that the last two terms add up to $\delta(D_h(u))$. This is not defined in the text but I think that we need to define $D_h u$ as the following

$$D_h u = D_h \sum_{j=1}^n F_j h_j(t) = \sum_{j=1}^n h_j(t) D_h(F_j) \tag{3.5} \quad \boxed{\text{P:3.2.4-4}}$$

Thus now $D_h(u)$ (3.5) is of the form $\sum F_j h_j$ and so we may apply the definition of δ to it to get the following

$$\delta(D_h u) = \sum_{j=1}^n B(h_j) D_h(F_j) - \sum_{j=1}^n \langle D_s(D_h F_j), h_j(s) \rangle \tag{3.6} \quad \boxed{\text{P:3.2.4-5}}$$

Now we focus in the last term of the right hand side of (3.6). From (3.4) we note that we want to show that

$$\sum_{j=1}^n \langle D_s(D_h F_j), h_j(s) \rangle = \sum_{j=1}^n \left\langle D_h (DF_j), h_j \right\rangle \tag{3.7} \quad \boxed{\text{P:3.2.4-7}}$$

Here are the steps,

$$\begin{aligned}
\sum_{j=1}^n \langle D_s(D_h F_j), h_j(s) \rangle &= \sum_{j=1}^n \left\langle D_s \left(\left\langle \sum_{i=1}^n \frac{\partial f^j}{\partial x_i} h_i^j(t), h(t) \right\rangle \right), h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle D_s \left(\sum_{i=1}^n \frac{\partial f^j}{\partial x_i} \langle h_i^j(t), h(t) \rangle \right), h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle \sum_{i=1}^n D_s \left(\frac{\partial f^j}{\partial x_i} \right) \langle h_i^j(t), h(t) \rangle, h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle \left\langle \sum_{i=1}^n D_s \left(\frac{\partial f^j}{\partial x_i} \right) h_i^j(t), h(t) \right\rangle, h_j(s) \right\rangle
\end{aligned}$$

$$\begin{aligned}
&= \sum_{j=1}^n \left\langle \left\langle \sum_{i=1}^n D_s \left(\frac{\partial f^j}{\partial x_i} h_i^j(t) \right), h(t) \right\rangle, h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle \left\langle D_s(D_t F_j), h(t) \right\rangle, h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle D_h(D F_j), h_j \right\rangle
\end{aligned}$$

Hence (3.7) is shown and we are done.

Proof of bottom of page 52

$$\begin{aligned}
\sum_{i,j=1}^{\infty} D_{e_i} \langle u, e_j \rangle D_{e_j} \langle v, e_i \rangle &= \sum_{i,j=1}^{\infty} \left\langle D_s \langle u, e_j \rangle, e_i(s) \right\rangle \left\langle D_t \langle v, e_i \rangle, e_j(t) \right\rangle \\
&= \sum_{i,j=1}^{\infty} \left\langle \langle D_s(u_t), e_j(t) \rangle, e_i(s) \right\rangle \left\langle \langle D_t(v_s), e_i(s) \rangle, e_j(t) \right\rangle \\
&= \sum_{i=1}^{\infty} \left\langle \sum_{j=1}^{\infty} \left(\langle D_s(u_t), e_j(t) \rangle \left\langle \langle D_t(v_s), e_i(s) \rangle, e_j(t) \right\rangle \right), e_i(s) \right\rangle \\
&= \sum_{i=1}^{\infty} \left\langle \left\langle D_s(u_t), \langle D_t(v_s), e_i(s) \rangle \right\rangle, e_i(s) \right\rangle \\
&= \sum_{i=1}^{\infty} \left\langle \left\langle D_s(u_t), \langle D_t(v_q), e_i(q) \rangle \right\rangle, e_i(s) \right\rangle \\
&= \sum_{i=1}^{\infty} \int_0^{\infty} \left(\int_0^{\infty} D_s(u_t) \left(\int_0^{\infty} D_t(v_q) e_i(q) dq \right) dt \right) e_i(s) ds \\
&= \int_0^{\infty} \left(\int_0^{\infty} D_s(u_t) \left(\sum_{i=1}^{\infty} \int_0^{\infty} D_t(v_q) e_i(q) dq e_i(s) \right) dt \right) ds \\
&= \int_0^{\infty} \int_0^{\infty} D_s(u_t) D_t(v_s) dt ds
\end{aligned}$$

Where above we use the identity

$$\langle f, g \rangle = \sum_{i=1}^{\infty} \langle f, e_i \rangle \langle g, e_i \rangle.$$

Proof of step from property 3

$$\begin{aligned}
E \left(\sum_{i=1}^{\infty} \langle v, e_i \rangle \left(\langle u, e_i \rangle + \delta(D_{e_i}(u)) \right) \right) &= E(\langle u, v \rangle) + E \left(\sum_{i=1}^{\infty} \langle v, e_i \rangle \delta(D_{e_i}(u)) \right) \\
&= E(\langle u, v \rangle) + \sum_{i=1}^{\infty} E \left(\left\langle D(\langle v, e_i \rangle), D_{e_i}(u) \right\rangle \right) \quad (3.8) \quad \boxed{\mathbf{s-1}} \\
&= E(\langle u, v \rangle) + \sum_{i=1}^{\infty} E \left(\left\langle \langle D_s(v_t), e_i(t) \rangle, \langle D_t(u_s), e_i(t) \rangle \right\rangle \right) \\
&= E(\langle u, v \rangle) + \sum_{i,j=1}^{\infty} E \left(\left\langle \langle D_s(v_t), e_i(t) \rangle, e_j(s) \right\rangle \left\langle \langle D_t(u_s), e_i(t) \rangle, e_j(s) \right\rangle \right)
\end{aligned}$$

$$\begin{aligned}
&= E(\langle u, v \rangle) + \sum_{i,j=1}^{\infty} E \left(D_{e_j} \langle v_t, e_i(t) \rangle \left\langle \langle D_t(u_s), e_j(s) \rangle, e_i(t) \right\rangle \right) \\
&= E(\langle u, v \rangle) + \sum_{i,j=1}^{\infty} E \left(D_{e_j} \langle v_t, e_i(t) \rangle D_{e_i} \langle u_s, e_j(s) \rangle \right) \\
&= E(\langle u, v \rangle) + \sum_{i,j=1}^{\infty} E \left(D_{e_j} \langle v, e_i \rangle D_{e_i} \langle u, e_j \rangle \right)
\end{aligned} \tag{3.9} \quad \boxed{\text{s-2}}$$

where in (3.8) we used Proposition 3.2.3 and in (3.9) we use Fubini's Theorem.

Nick's Question 3.4 (Why is δ linear?) On page 55 it says that δ is linear. First, for $u = \sum_{j=1}^n F_j h_j$ we have

$$\delta(u) = \sum_{j=1}^n F_j B(h_j) - \sum_{j=1}^n \langle DF_j, h_j \rangle_H$$

Now if we have a $v = \sum_{j=1}^n G_j h'_j$ how does $\delta(u + v) = \delta(u) + \delta(v)$. Also how could we write $\sum_{j=1}^n F_j h_j + \sum_{j=1}^n G_j h'_j = \sum_{j=1}^n K_j k_j$ where $K_j \in \mathcal{S}$ and $k_j \in H$

Do we do this through the relationship $E(\langle DF, u \rangle) = E(\delta(u)F)$ in the sense that

$$\begin{aligned}
E(\delta(u + v)F) &= E(\langle DF, u + v \rangle) \\
&= E(\langle DF, u \rangle) + E(\langle DF, v \rangle) \\
&= E(\delta(u)F) + E(\delta(v)F) \\
&= E((\delta(u) + \delta(v))F)
\end{aligned}$$

and if this is true for every F then I believe we can say $\delta(u + v) = \delta(u) + \delta(v)$?

Nick's Question 3.5 Exercise 3.1 I don't see how we can apply this hint? Suppose we are able to show that the hint is true then how does that show $\mathcal{P}$ is dense in $L^q(\Omega)$? There is a theorem that $\{x_n\}$ is a basis for a Hilbert space H if and only if $\sum \langle x, x_n \rangle x_n = 0$ implies $x = 0$ but this is only valid for Hilbert space and $L^q(\Omega)$ for $q \neq 2$ is not a Hilbert space since there is no inner product.

Nick's Question 3.6 Prop 3.3.2 page 54 This relates to exercise 3. In the proof of prop 3.3.2 how can we use Holders inequality to show that $\varphi'(F)D(F) \in L^q(\Omega, H)$?

Nicks Solution.

$$\begin{aligned}
[E \|\varphi'(F)DF\|_H^q]^{1/q} &= \left[\int_{\Omega} \left(\int_0^{\infty} \varphi'(F)^2 (D_t F)^2 dt \right)^{q/2} dP \right]^{1/q} \\
&\leq \left[\int_{\Omega} (1 + |F|^\alpha)^q \left(\int_0^{\infty} (D_t F)^2 dt \right)^{q/2} dP \right]^{1/q} \\
&= \left\| (1 + |F|^\alpha) \left(\int_0^{\infty} (D_t F)^2 dt \right)^{1/2} \right\|_q
\end{aligned}$$

Note that $\frac{1}{q} = \frac{1}{p} + \frac{1}{p/\alpha}$ and so if we apply Holders Inequality above we get that

$$[E \|\varphi'(F)DF\|_H^q]^{1/q} \leq \|1 + |F|^\alpha\|_{p/\alpha} \left\| \left(\int_0^\infty (D_t F)^2 dt \right)^{1/2} \right\|_p$$

which is finite since $F \in \mathbb{D}^{1,p}$. □

Nick's Question 3.7 Prop 3.3.2 page 54

I have not been able to verify why $\varphi(F) \in \mathbb{D}^{1,q}$. By definition, this means that we need to show that there exists $X_n \in \mathcal{S}$ such that

1. $X_n \rightarrow \varphi(F)$ in $L^q(\Omega)$
2. $D(X_n) \rightarrow D(\varphi(F))$ in $L^q(\Omega, H)$

By assumption we already have that there exists $\{F_n\} \subset \mathcal{S}$ such that $F_n \rightarrow F$ in $L^p(\Omega)$ and also $DF_n \rightarrow DF$ in $L^p(\Omega, H)$. We then take a sequence of mollifiers, $\{\alpha_n\}$ and consider the functions $\varphi_k(x) = (\varphi * \alpha_k)(x)$ which by the original assumption on φ are $\alpha + 1$ holder continuous.

Now, we notice that by properties of the mollifier since $\varphi(F) \in L^q(\Omega)$ then

$$E|\varphi_k(F_j) - \varphi(F_j)|^q \xrightarrow{j \rightarrow \infty} 0$$

Next, since φ is $\alpha + 1$ holder continuous then we have

$$\begin{aligned} E|\varphi(F_j) - \varphi(F)|^q &\leq KE|F_j - F|^{q(\alpha+1)} \\ &= KE|F_j - F|^p \rightarrow 0 \end{aligned}$$

and thus we conclude as k and j go to infinity that $\varphi_k(F_j) \rightarrow \varphi(F)$ in $L^q(\Omega)$ and $\varphi_k(F_j) \in \mathcal{S}$.

I am unable to show why $D_t(\varphi_k(F_j)) \rightarrow D_t(\varphi(F))$ in $L^q(\Omega, H)$.

However since $F_j = f^j(B(h_1^j), \dots, B(h_n^j))$, we see that

$$\begin{aligned} D_t(\varphi_k(F_j)) &= \sum_{i=1}^n \frac{\partial}{\partial x_i} (\varphi * \alpha_k)(F_j) h_i^j(t) \\ &= \sum_{i=1}^n (\varphi' * \alpha_k)(F_j) \frac{\partial f^j}{\partial x_i} h_i^j(t) \\ &= (\varphi' * \alpha_k)(F_j) \sum_{i=1}^n \frac{\partial f^j}{\partial x_i} h_i^j(t) \\ &= (\varphi' * \alpha_k)(F_j) D_t(F_j). \end{aligned}$$

We know that $\{\alpha_k\}$ are mollifiers so $(\varphi' * \alpha_k)(F_j) \rightarrow \varphi'(F_j)$ almost surely and by assumption $D_t(F_j) \rightarrow D_t(F)$ in $L^p(\Omega, H)$. But I still am having trouble rigorously showing that $D_t(\varphi_k(F_j)) \rightarrow D_t(\varphi(F))$ in $L^p(\Omega, H)$.

Nick's Question 3.8 Exercise 3.4

Let $M = \sum_{t \in [0,1]} X_t$. The hint says to approximate this by taking the sup over a finite set. We know that since X_t is continuous in t then $M = \sup_{t \in \mathbb{A}} X_t$ where $A = \mathbb{Q} \cap [0, 1]$. Let $\{a_n\}$ be an enumeration of $A \setminus \{0, 1\}$ and let $S_k = \{0, a_1, \dots, a_k, 1\}$ then each S_k are finite and $M_k := \sup_{t \in S_k} X_t$ is an approximation

of M since $M_k \rightarrow M$ almost surely. The hint also tells us to use Corollary 4.2.5 but that requires us to find $D_t M_k$ which I am unsure of how to do. The hint says that we need to use exercise 3.3 which is chain rule for Lipschitz functions but I do not know how that applies. This corollary would also require us to show that $M_k \xrightarrow{L^2(\Omega)} M$ which I have been unable to show either.

3.2 Exercises

Exercise 3.3 Let α_n be a sequence of mollifiers and $\varphi_k := (\varphi * \alpha_k)$. Since φ is Lipschitz with Lipschitz constant K then this implies that $\varphi'_k \leq K$ and in addition φ' exists almost everywhere and is also bounded by K . By proposition 3.3.2 we have that

$$D_t(\varphi_k(F)) = \varphi'_k(F)D_t(F)$$

Also note that

$$\begin{aligned} |\varphi_k(F) - \varphi(F)|^2 &= \left| \int_{\mathbb{R}} (\varphi(F - y) - \varphi(F)) \alpha_k(y) dy \right|^2 \\ &\leq K \left| \int_{\mathbb{R}} y \alpha_k(y) dy \right|^2 \\ &\leq K' \end{aligned}$$

Hence $|\varphi_k(F) - \varphi(F)|^2$ is bounded above and also converges to 0 almost surely so by the dominated convergence theorem we have that

$$\mathbb{E}|\varphi_k(F) - \varphi(F)|^2 \rightarrow 0$$

and so the hypothesis of corollary 4.2.5 is satisfied. This implies that $\varphi(F) \in \mathbb{D}^{1,2}$ and

$$D_t(\varphi_n(F)) \rightarrow D_t(\varphi(F))$$

weakly in $L^2(\Omega, H)$. Lastly, since $\varphi'_k(F)D_t(F)$ is bounded in $L^2(\Omega, H)$ and also

$$\varphi'_k(F)D_t(F) \rightarrow GD_t(F)$$

almost surely where G is a random variable bounded by K then

$$\varphi'_k(F)D_t(F) \rightarrow GD_t(F)$$

weakly in $L^2(\Omega, H)$ as well. Since weak limits are unique then we must have

$$D_t(\varphi(F)) = GD_t(F)$$

However note that in general we have as a property of mollifiers that $\varphi'_n(x) \rightarrow \varphi'(x)$ almost surely. But we said before that $\varphi'(x)$ only exists almost surely since φ is Lipschitz. Therefore if the law of F is absolutely continuous with respect to Lebesgue measure then we have $G = \varphi'(F)$ almost surely. Indeed, let $\mathcal{N}$ be the null set where φ' is undefined. If the law of F is absolutely continuous with respect to Lebesgue measure then $P(F \in \mathcal{N}) = 0$ and we must have $G = \varphi'(F)$ almost surely.